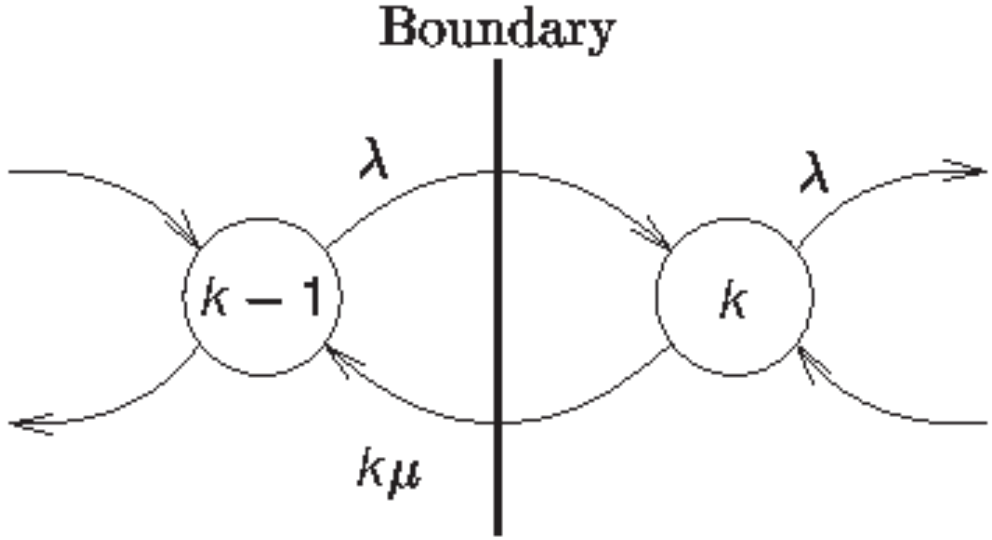
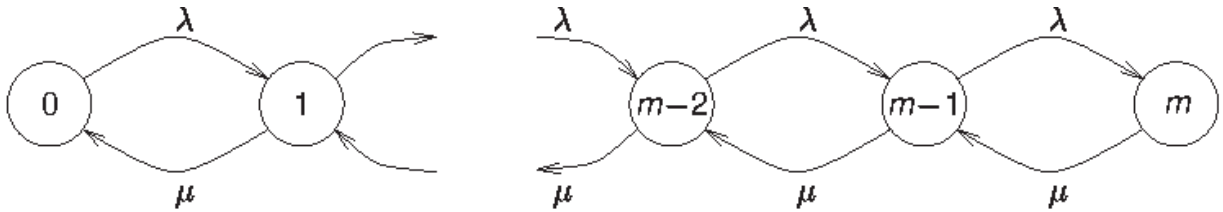
Other Markov Queues

2006-10-16

State diagram for the $M/M/\infty$ queue.



Flow rates at a boundary for $M/M/\infty$ queue.



State diagram for the $M/M/1/m$ queue. Note that there are no transitions to states above the mth state.



Other Markov Queues

- The $M/M/1$ queue is perhaps the simplest Markov queue.
- We will introduce two related Markov queues and compute their stationary distributions.

Other Markov Queues (continued)

- A simple modification to the $M/M/1$ queue: a *finite length*modification $M/M/1/m$
- The *infinite server*queue,$M/M/\infty$.

Finite Length $M/M/1/m$



stateMM1m

Finite Length $M/M/1/m$ (continued)

- We have assumed that $M/M/1$ queue can have an infinite length.
- In practice a restricted buffer size — perhaps queue length is also restricted.
- We assume the system can contain at maximum m customers (*i.e.*the queue can contain $m-1$).
-

Examples

- Example 1: Derive the stationary distribution for this queue.
- Example 2: Blocking probabilities

Infinite Serving Capacity — $M/M/\infty$

- The $M/M/1/m$ queue covers the case where the queue length is bounded.
- We can also derive when the number of servers is *unbounded*.
- In the $M/M/1$ queue we assume that there is only one server.

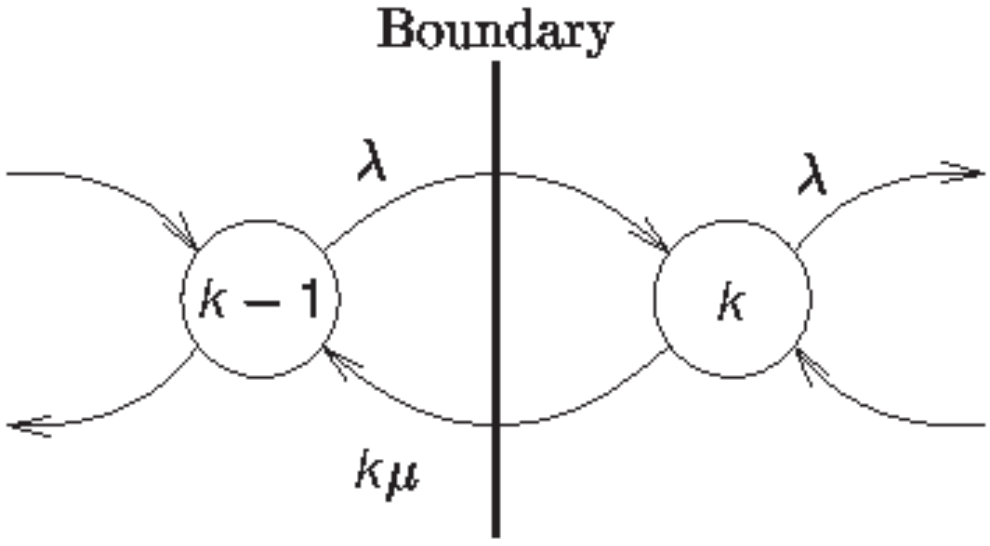
Infinite Serving Capacity — $M/M/\infty$ (continued)

- It can be useful to consider an unlimited number of servers:
- One server for every customer in the system.
- The state diagram for this system can be drawn.



stateMMinf

Detailed Balance



boundaryFlowMMinf

Detailed Balance (continued)

- Make use of the standard detailed balance equations for this queue.
-
- From the flow rates at the boundary we have in general

$$\lambda_{k-1} P(k-1) = \mu_k P(k)$$

- There is infinite service capacity in the queue the service rate at state k is simply $\mu_k = k\mu$.
- The rate of arrivals stays constant, $\lambda_k = \lambda$, so for this queue detailed balance gives.

$$\lambda P(k-1) = k\mu P(k)$$

- Reorganising in terms of $P(k)$ then gives

$$P(k) = \frac{\lambda}{k\mu} P(k-1)$$

Normalisation Condition

- Using the normalisation condition for this queue,

$$\sum_{k=0}^{\infty} P(k) = 1,$$

we obtain $P(0)$ as

$$P(0) = \left[\sum_{k=0}^{\infty} \left(\frac{\lambda}{\mu} \right)^k \frac{1}{k!} \right]^{-1}.$$

The Offered Load

- It is convenient to define the offered load,

$$a = \frac{\lambda}{\mu}$$

as arrival rate divided by service rate. * This is a dimensionless constant, but has international standard units of Erlangs (after A. K. Erlang).

The Offered Load (continued)

- Substituting this definition back into (12) we have,

$$P(0) = \left[\sum_{k=0}^{\infty} \frac{a^k}{k!}\right]^{-1}.$$

- Recalling that

$$\sum_{n=0}^{\infty} \frac{x^n}{n!} = \exp(x)$$

we have

$$P(0) = \exp(-a)$$

and the steady state distribution is then

$$P(k) = \frac{a^k}{} \exp(-a)$$

Utilization Factor

- Notice that the utilization factor for this system, ρ, doesn't really exist.
- For the $M/M/\infty$ there is infinite capacity so we can't ask what percentage is in use!
- The proporation of time that some part of the system is in use is

$$1 - P(0) = 1 - \exp(-a).$$

Thanks!

- company: Trent AI
- book: The Atomic Human
- twitter: @lawrennd
- The Atomic Human
- newspaper: Guardian Profile Page
- blog: http://inverseprobability.com

Further Reading

- Birth-death queues beyond $M/M/1$: finite storage and infinite servers of Bertsekas and Gallager (1992)

References

Bertsekas, D.P., Gallager, R.G., 1992. Data networks, Second. ed. Prentice-Hall, Upper Saddle River, NJ.